Data Quality Report

NexusMart Datathon | Final cleaning outputs and reconciliation evidence | 18 October 2026

1. Scope and method

Nine source CSVs were profiled, standardized and validated using reusable Python cleaning functions. The outputs include cleaned datasets, a consolidated rejects file, customer/product ID maps and step-level accounting. Apache Airflow orchestrates extraction, raw validation, cleaning, warehouse loading, dimensional/fact construction and checks. This report distinguishes measured issues, repairs and outstanding review items; it does not certify unprovided warehouse verification results.

2. Source-level before/after counts

Dataset	Raw rows	Clean rows (local stage)	Removed / merged
locations	650	650	0
sellers	420	420	0
products	3,200	3,096	104
customers	45,000	42,983	2,017
orders	163,200	160,000	3,200
order_items	315,143	315,143	0
payments	168,111	164,815	3,296
shipping	151,228	151,228	0
returns	16,048	16,048	0

At every one of the 29 recorded processing stages, rows_in = rows_out + rows_rejected. These stages include sequential dependency checks; their counts must not be summed as independent raw datasets. The final 8,820-row rejects file is the reliable consolidated total for rejected/merged records.

3. Principal cleaning outcomes

Issue	Evidence / treatment
Customers merged	2,017 customer identifiers mapped to surviving master IDs; customer_map.csv contains 2,017 mappings.
Products merged	104 product identifiers mapped to masters; product_map.csv contains 104 mappings.
Exact duplicates	3,200 orders and 3,296 payments removed and logged as EXACT_DUPLICATE.
Order dates	1,593 order dates repaired; 12 unreparable order-date records rejected.
Broken references	321 customer references and 932 product references mapped to Unknown; 29 orphan order items, 12 orphan payments, 5 orphan shipments and 140 orphan returns rejected.
Prices / discounts	300 unit prices scaled down by 100; 308 USD-valued prices converted; 947 prices repaired from list price; 5 invalid-priced items rejected. 31,565 missing discounts recomputed.
Shipping fees	9,477 fees filled using the documented typical-fee rule; these are imputations, not observed fees.
Order totals	1,016 order IDs flagged because sum of line net amounts differs from recorded order total.

4. Rejected-record traceability

The final rejects.csv contains 8,820 records. Each contains source_file, record_id, reason_code, detail, and raw_record, supporting traceability rather than silent row removal.

Reason code	Records
EXACT_DUPLICATE	6,496
MERGED_DUPLICATE	2,121

ORPHAN_ORDER_ITEM	140
ORPHAN_ORDER	46
ORDER_DATE_UNREPAIRABLE	12
INVALID_PRICE	5

5. Reconciliation (latest uploaded CSV)

The uploaded metric_reconciliation.csv compares independently computed Python and SQL headline metrics. The percentage rate is reported on a 0–100 percentage scale.

Metric	Python	SQL	Difference	Result
Total orders	159,988.00	159,988.00	0.000000	MATCH
Net revenue	1,141,271,695.00	1,141,271,695.00	0.000000	MATCH
Refund amount	58,033,099.00	58,033,099.00	0.000000	MATCH
Return rate	5.148291	5.148300	0.000009	MATCH
Shipping fee total	81,388,942.00	81,388,942.00	0.000000	MATCH

All five comparisons are marked MATCH in the supplied file. Power BI has not yet been included; three-layer reconciliation remains pending. IMPORTANT: the uploaded reconciliation CSV reports Net revenue = PKR 1,141,271,695, whereas an earlier SQL screenshot in this conversation showed PKR 11,412,716,195. Confirm the final correct warehouse run and regenerate the comparison if necessary before submission.

6. Known limitations and checks pending

- Unknown dimension references are retained to avoid losing sales; report their actual final FactSales counts from verify_star_v2.sql.
- Shipping-fee imputation (9,477 records) and 1,016 order-total discrepancies may affect financial interpretation; do not assume all differences are rounding.
- Legacy profiling artifacts dq_rejects.csv and dq_review.csv represent audit-stage observations, not the finalized 8,820 production rejects; do not add their row counts to production removals.
- Warehouse verify_star_v2.sql PASS / FAIL / REVIEW output has not been supplied here. Its outcomes must be recorded after running it, especially shipping allocation, return totals, Unknown keys and stale dimensions.
- Power BI headline cards and unfiltered DAX metrics must still be checked against finalized Python/SQL results.

7. Evidence and conclusion

Evidence files: clean_stats(1).csv; rejects(1).csv; customer_map(1).csv; product_map(1).csv; dq_log(1).csv; dq_rejects.csv; dq_review.csv; metric_reconciliation.csv. The consolidated cleaning process maintains row accounting and reason-coded rejections; customer/product merges preserve cross-file mappings. Five uploaded Python/SQL metrics match numerically, subject to the reported historical revenue discrepancy and pending Power BI/warehouse check outputs.